

ECE 3441 Spring 2025

Quiz 1

VERSION A

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 60 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

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SOLUTIONS* *2025/02/24*
First Name Last Name, Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		30
2		20
3		20
4		15
5		25

Total: _____/(100 possible points + 10 bonus points)

Question #1:

- a) Convert the following decimal number to binary (10 points).

$$125.4375_{10}$$

$$\begin{array}{rcl} 125 : 1 & 0.4375 \\ 62 : 0 & 0.8750 \\ 31 : 1 & 1.7500 \\ 15 : 1 & 1.5000 \\ 7 : 1 & 1.0000 \\ 3 : 1 & \\ 1 : 1 & \\ 0 & \end{array}$$

$$\Rightarrow 111, 1101.0111_2$$

- b) Convert the value in a) to hexadecimal (two digits after the radix point). (5 points)

$$\underbrace{0111}_7, \underbrace{1101}_{13}, \underbrace{0111}_7, \underbrace{0000}_0$$

$$\Rightarrow 7D.70_{16}$$

- c) Using **8-bit 2's-complement** binary arithmetic, perform the subtraction below and convert the answer to decimal. Remember: first convert each decimal number to its 8-bit 2's-complement binary representation before performing the operation. (15 points)

$$57_{10} - 12_{10}$$

$$\begin{array}{rcl} 57 : 1 \\ 28 : 0 \\ 14 : 0 \\ 7 : 1 \\ 3 : 1 \\ 1 : 1 \\ 0 \end{array}$$

$$57_{10} \Rightarrow 0011, 1001_2$$

$$-12_{10} \Rightarrow 0000, 1100$$

$$\hookrightarrow 1111, 0011$$

$$\hookrightarrow 1111, 0100_2$$

$$\begin{array}{r} 0011, 1001 \\ + 1111, 0100 \\ \hline \end{array}$$

$$1,0010, 1101$$

$$\Rightarrow 0010, 1101_2$$

$$= 1 + 4 + 8 + 32 = 45_{10}$$

$$\begin{array}{rcl} 12 : 0 \\ 6 : 0 \\ 3 : 1 \\ 1 : 1 \\ 0 \end{array}$$

Question #2:

For each expression, identify what theorem/law can be used to simplify it **in one step**. Then, apply the theorem to the expression. Remember: you can treat expressions in parentheses like whole variables in the theorems. (20 points)

a) $XY'Z + ZX$

$$= XZ \quad : \text{Absorption (2-3, 2)}$$

b) $XYZ + Y'Z$

$$= XZ + Y'Z \quad : \text{Elimination (2-3, 3)}$$

c) $(ABC' + D)(ABC' + EF')$

$$= ABC' + DEF' \quad : \text{Distributive (2-4, 8D)}$$

d) $ABC'E' + B'ADC' + EDC'A$

~~$$= ABC'E' + B'ADC' + EDC'A + AC'DE' + AC'BD \quad : \text{Consensus (2-4, 8D)}^6$$~~

$$= AC'(BE' + B'D + DE) \quad : \text{Distributive}$$

$$= AC'(BE' + B'D + DE + DE') \quad : \text{Consensus}$$

$$= AC'(BE' + B'D + D) \quad : \text{Uniting}$$

$$= AC'(BE' + D) \quad : \text{Absorption}$$

Question #3:

- a) Eliminate the XOR and then factor to obtain the minimum POS of the following. (10 points)

Let $F = (AB \oplus C) + C'D$

$$F = ABC' + A'C + B'C + C'D$$

~~$$F = F'' = ((A' + B' + C)(A + C')(B + C')(C + D'))^3$$~~

~~$$= (((A' + B')D' + C)(AB + C'))^3$$~~

~~$$= ((A'D' + B'D' + C)(AB + C'))^3$$~~

~~$$= (A'C'D' + B'C'D' + ABC)^3$$~~

~~$$= (A + C + D)(B + C + D)(A' + B' + C')$$~~

AB \ CD	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	1	1	0	1
10	1	1	0	1

POS:

$$(A + C + D)(B + C + D)(A' + B' + C')$$

* Redone at end. *
on next page.

- b) Prove or disprove the following: $A \oplus BC = (A \oplus B)(A \oplus C)$. (10 points)

$$A \oplus BC = A'(BC) + A(BC)'$$

$$= A'BC + AB' + AC'$$

~~$$(A \oplus B)(A \oplus C) = (AB' + A'B)(AC' + A'C)$$~~

$$= AB'C' + A'BC$$

A \ BC	00	01	11	10
0	0	0	1	0
1	1	1	0	1

A \ BC	00	01	11	10
0	0	0	1	0
1	1	0	0	0

$$A'BC + AB' + AC' \neq AB'C' + A'BC$$

$$\therefore A \oplus BC \neq (A \oplus B)(A \oplus C)$$

* Any counterexample would also work *

Question #4:

Simplify the following to only 3 terms using only the consensus theorem or its dual. (15 points)

$$a) A'BC' + BC'D' + \underline{A'CD} + \underline{B'CD} + \underline{A'BD}$$

$$= \underline{A'BC'} + \underline{BC'D'} + B'CD + \underline{A'BD}$$

$$= \boxed{BC'D' + B'CD + A'BD}$$

/ * continued from earlier * /

$$3.) a.) (AB \oplus C) + C'D$$

$$= ABC' + (AB)'C + C'D \quad : \text{XOR}$$

$$= ABC' + (A' + B')C + C'D \quad : \text{DeMorgan's}$$

$$\equiv \cancel{ABC'} + \cancel{A'C} +$$

$$= (A' + B')C + (AB + D)C' \quad : \text{Distributive}$$

$$= (A' + B')C + (A + D)(B + D)C' \quad : \text{MOAIF Distributive}$$

$$= ((A + D)(B + D) + C)(A' + B' + C') \quad : \text{MOAIF}$$

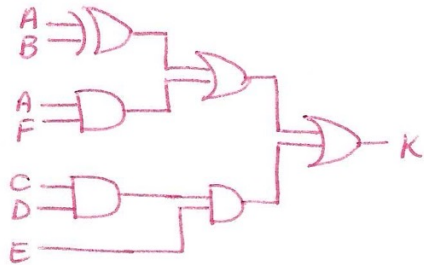
$$= (A + C + D)(B + C + D)(A' + B' + C') \quad : \text{Distributive.}$$

Question #5:

- a) Simplify and sketch a circuit that uses no more than six 2-input logic gates (10 points).

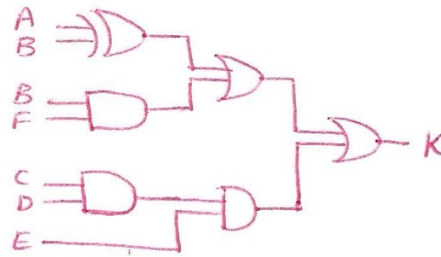
$$K = AB' + \underline{A'B} + \underline{AF} + \underline{BF} + CDE$$

$$\begin{aligned} K &= AB' + \underline{A'B} + \underline{AF} + \underline{BF} + CDE \\ &= AB' + A'B + AF + CDE \\ &= A \oplus B + AF + CDE \end{aligned}$$



circuit 1

$$\begin{aligned} K &= \underline{AB'} + A'B + \underline{AF} + \underline{BF} + CDE \\ &= AB' + A'B + BF + CDE \\ &= A \oplus B + BF + CDE \end{aligned}$$



circuit 2

** XOR will not be required. Getting to 7 or 8 gates is fine **

- b) For the circuit implemented in Part a), calculate the maximum pull-up resistance that can be used for input B, using the provided data (15 points).

Table 1: Input and Output Parameters for Various TTL Chips

		74LS00 (NAND)	74LS02 (NOR)	74LS04 (NOT)	74LS08 (AND)	74LS32 (OR)	74LS86 (XOR)
V _{CC}	NOM	5 [V]					
V _{IH}	MIN	2 [V]	2 [V]	2 [V]	2 [V]	2 [V]	2 [V]
V _{IL}	MAX	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]
V _{OH}	MIN	2.5 [V]	2.4 [V]	2.4 [V]	2.7 [V]	2.7 [V]	2.7 [V]
	TYP	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]
V _{OL}	TYP	0.35 [V]	0.2 [V]	0.2 [V]	0.35 [V]	0.35 [V]	0.35 [V]
	MAX	0.5 [V]	0.4 [V]	0.4 [V]	0.5 [V]	0.5 [V]	0.5 [V]
I _{OH}	MAX	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]
I _{OL}	MAX	8 [mA]	16 [mA]	16 [mA]	8 [mA]	8 [mA]	8 [mA]
I _{IH}	MAX	20 [μA]	40 [μA]	20 [μA]	20 [μA]	20 [μA]	40 [μA]
I _{IL}	MAX	-0.4 [mA]	-1.6 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.8 [mA]

/* Treat B independent of A; Take their statement and ~~run~~ run with it. */

SCRATCH

B is fed to XOR and AND

$$\text{XOR: } R_{PU} < \left(\frac{(5[V]) - (2[V])}{(40 [\mu A])} = \boxed{75 [k\Omega]} \right)$$

$$\text{AND: } R_{PU} < \left(\frac{(5[V]) - (2[V])}{(20 [\mu A])} = 150 [k\Omega] \right)$$

/* If multiple values are calculated then they should choose the lowest */

* Circuit 1 needs only XOR

Circuit 2 needs both XOR and AND

Uniting theorems:

1. $XY + XY' = X$

Absorption theorems:

2. $X + XY = X$

Elimination theorems:

3. $X + X'Y = X + Y$

Duality:

4. $(X + Y + Z + \dots)^D = XYZ \dots$

Theorems for multiplying out and factoring:

5. $(X + Y)(X' + Z) = XZ + X'Y$

Consensus theorems:

6. $XY + YZ + X'Z = XY + X'Z$

Operations with 0 and 1:

1. $X + 0 = X$

2. $X + 1 = 1$

Idempotent laws:

3. $X + X = X$

Involution law:

4. $(X')' = X$

Laws of complementarity:

5. $X + X' = 1$

Commutative laws:

6. $X + Y = Y + X$

Associative laws:

7. $(X + Y) + Z = X + (Y + Z)$
 $= X + Y + Z$

Distributive laws:

8. $X(Y + Z) = XY + XZ$

DeMorgan's laws:

9. $(X + Y)' = X'Y'$

Exclusive OR

$X \oplus Y = X'Y + XY'$

1D. $(X + Y)(X + Y') = X$

2D. $X(X + Y) = X$

3D. $X(X' + Y) = XY$

4D. $(XYZ \dots)^D = X + Y + Z + \dots$

5D. $XY + X'Z = (X + Z)(X' + Y)$

6D. $(X + Y)(Y + Z)(X' + Z) = (X + Y)(X' + Z)$

1D. $X \cdot 1 = X$

2D. $X \cdot 0 = 0$

3D. $X \cdot X = X$

5D. $X \cdot X' = 0$

6D. $XY = YX$

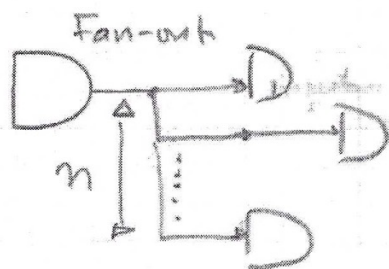
7D. $(XY)Z = X(YZ) = XYZ$

8D. $X + YZ = (X + Y)(X + Z)$

9D. $(XY)' = X' + Y'$

Equivalence

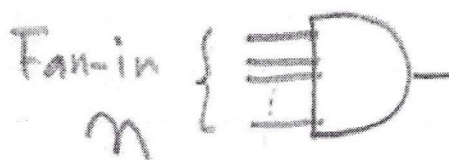
$(X \equiv Y) = XY + X'Y'$



Fan-out $L = n$

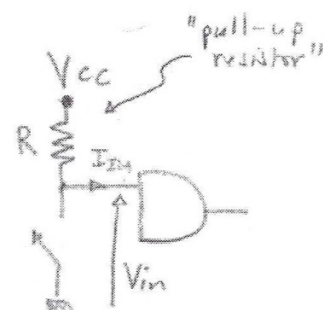
Fan-out $H = m$

Fan-out $= \min(m, n)$



$m < \frac{I_{OHmax}}{I_{IHmax}}$

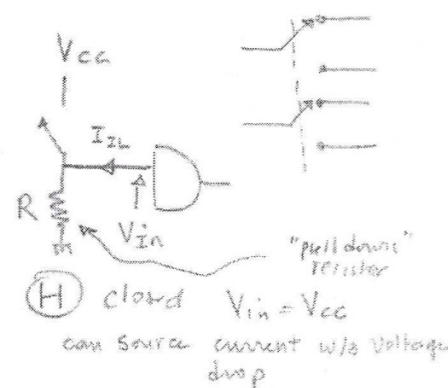
$n < \frac{I_{OLmax}}{I_{ILmax}}$



(L) closed $V_{in} = 0$ (L)
 can sink current without generating voltage

(H) open
 $V_{in} = V_{cc} - I_{IH}R$
 $V_{in} > V_{IHmin}$

$V_{cc} - I_{IH}R > V_{IHmin}$
 $\boxed{\frac{V_{cc} - V_{IHmin}}{I_{IHmax}} > R}$



(H) closed $V_{in} = V_{cc}$
 can source current w/o voltage drop

(L) open
 $V_{in} = RI_{IL}$
 $V_{in} < V_{ILmax}$

$RI_{IL} < V_{ILmax}$

$\boxed{R < \frac{V_{ILmax}}{I_{ILmax}}}$

